

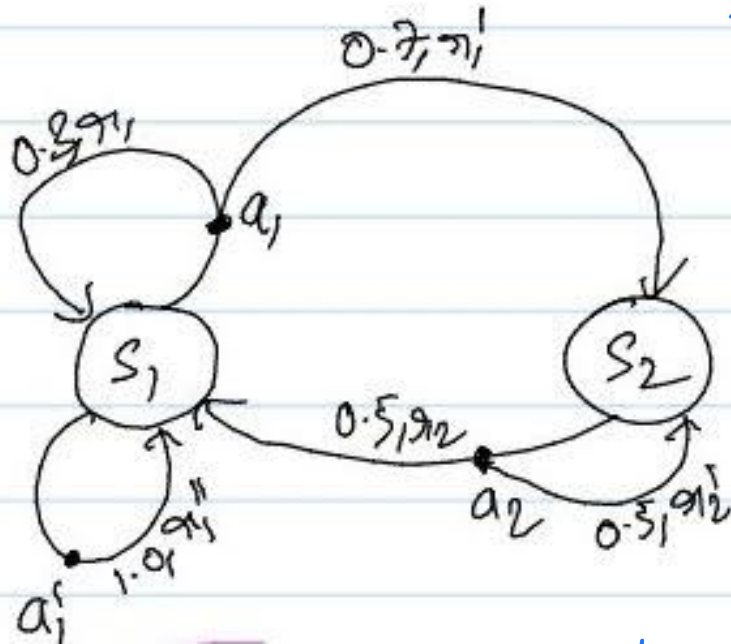
Capturing The Stochastic (Markov) Environment

- The Markov Property
 - The future is entirely determined by the current state and your action
 - Given the current, no amount of information from the past is beneficial
- MDP is just a bunch of probability mass functions (PMFs)

$$P[S_{t+1} = s', R_{t+1} = r | S_t = s, A_t = a],$$

for every (s, a) .

MDP Example (Graphical Representation)



$$P[S_{t+1}=s', R_{t+1}=r | S_t=s, A_t=a]$$

$$= \begin{cases} 0.7 & (s'=s_2, r=r_1') \\ 0.3 & (s'=s_1, r=r_1) \\ 0 & \text{otherwise} \end{cases}$$

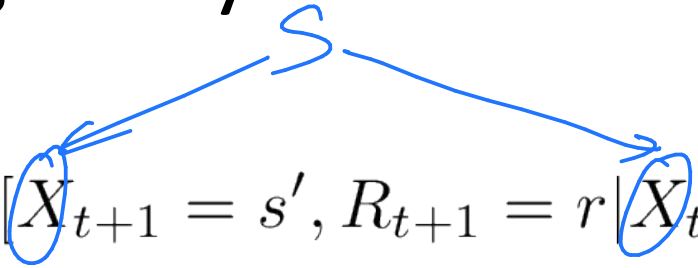
Conditional joint PMF of S_{t+1}, R_{t+1} , given $S_t=s, A_t=a$.

Similarly I have joint PMF(s):

$$P[S_{t+1}=s', R_{t+1}=r | S_t=s, A_t=a] = \begin{cases} P[S_{t+1}=s', R_{t+1}=r | S_t=s_1, A_t=a_1] \\ P[S_{t+1}=s', R_{t+1}=r | S_t=s_2, A_t=a_2] \end{cases}$$

MDP doesn't guide choice of actions.
It does constrain the set of allowed actions in a given state.

Time Homogeneity

$$p(s', r | s, a) \triangleq P[X_{t+1} = s', R_{t+1} = r | X_t = s, A_t = a]$$


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$$p(s', r | s, a) \triangleq P[X_{t+1} = s', R_{t+1} = r | X_t = s, A_t = a]$$

$$r(s, a) \triangleq E[R_{t+1} | X_t = s, A_t = a]$$

Time Homogeneity

$$p(s', r | s, a) \triangleq P[X_{t+1} = s', R_{t+1} = r | X_t = s, A_t = a]$$

$$\sum_{r, s'} \pi p(s', \pi | s, a)$$

$$r(s, a) \triangleq E[R_{t+1} | X_t = s, A_t = a] = \sum_{\pi} \pi \sum_{s'} p(s', \pi | s, a)$$

$$\begin{aligned} r(s, a, s') &\triangleq E[R_{t+1} | X_t = s, A_t = a, X_{t+1} = s'] \\ &= \sum_{\pi} \pi \frac{P[X_{t+1} = s', R_{t+1} = \pi | X_t = s, A_t = a]}{P[X_{t+1} = s' | X_t = s, A_t = a]} \end{aligned}$$

Setting Rewards

Return G_t

- Episodic task

$$G_t = R_{t+1} + R_{t+2} + \dots + R_T,$$

Where T is a random variable which takes finite values v.p. ∞ .

G_t is not known, that is the value that G_t takes is only known at the end of the episode.

Return

→ Is $P(R_{t+2}=x | S_t=s, S_{t+1}=s')$ equal/not to $P(R_{t+2}=x | S_{t+1}=s')$

If X, Y
are i.i.d.

then
 $P(X=x | Y=y)$
 $= P(X=x)$
for all x, y .

- Continuing task

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots, \quad 0 \leq \gamma < 1.$$

Is R_{t+2} dependent on S_t ? Yes.

Is R_{t+2} dependent on S_t , given A_t ?

Is R_{t+2} dependent on S_t , given S_{t+2} ?

Is R_{t+2} dependent on S_t , given S_{t+1} ? No.

Is R_{t+2} dependent on S_t , given S_{t+1} ? No.

Policy

Deterministic:

$$\pi: S \rightarrow A$$

(S is the set of all states)

$\pi(s)$ is the action one takes in state s when using policy π .

Stochastic:

Policy π is a conditional PMF of actions, given the state.

$$\pi(a|s) = P[A_t = a | S_t = s]$$

One PMF for every s .

for every action a in state s .

State-Value Function

$$V_{\pi}(s) \rightarrow \mathbb{R}$$



- A function that maps states to their **values**
- The **value** of a state is simply the expected return obtained starting in the state

$$V_{\pi}(s) = E[G_t | S_t = s]$$

- Clearly, it depends on the policy used by the agent

$S_0=0, S_1=0, S_2=1, S_3=0, S_4=1, S_5=0, S_6=1, S_7=T$
 $R_0=1, R_1=2, R_2=-1, R_3=0, R_4=6, R_5=7, R_6=0$
 $E[G_t | S_t = 0] = ?$

State-Value Function

$$v_{\pi}(s) = E_{\pi}[G_t | S_t = s]$$

Action-Value Function

$$q_{\pi}(s, a) = E_{\pi}[G_t | S_t = s, A_t = a]$$

It is used to select actions $A_{t+1}, A_{t+2}, \dots$

Must assign a value for every (s, a) in the cartesian product of the set of states $\rightarrow$ the set of actions.

State-Value in Terms of Action-Value?

$V_{\pi}(s)$ in terms of $Q_{\pi}(s,a)$

- Deterministic Policy $V_{\pi}(s) = Q_{\pi}(s, \pi(s))$
- Randomized/Stochastic Policy

$$\hookrightarrow V_{\pi}(s) = \sum_a Q_{\pi}(s,a) \pi(a|s)$$

$$\sum_a Q_{\pi}(s,a) P\{A_t = a | S_t = s\}$$